Lec 3:

التسلسل العام بيكون =

(A) Pharmacist in public health

1- Puplic Health definition ← تعريف

2- 4 levels of participation ← الاربع مستويات

3- General ① roles + Rational ② * ← ادوار الصيدلي (العامة ① + الأساسية ②)

4- Applications / Examples

- Vector-born disease
- viral infection
- Bacterial infection
- TB
- Disaster management
- De-addiction
- Maternal health
- Global health
- Chronic diseases

(B) pharmacy Job opportunities

5- Community → Hospital → clinical → Industrial

Industrial → R&D → Manufacturing → QC

QC → Quality control

1) Definition

Public Health = art + science of =

3P = Prevent → Prolong → Promote

EHR = Education → Healthy lifestyle → Research ← through منخلال

2) 4 Levels :

Levels	Main Idea	باختصر
Primordial	Preventing of Risk factors	No patient ماشي مريض
Primary	Preventing of disease occurrance	
Secondary	Reduction of disease impact	patient بيكون في مريض
Tertiary	Reduction of long-term impact	

من باب الربط :

Primordial → Risk

Primary → Disease

Secondary → impact

Tertiary → long-term impact

3) Pharmacist in Public Health

General Roles العام

- Campaigns حملات
- National programs برامج وطنية
- De-addiction programs برامج فك ادمان
- Dispensing صرف / patinent counseling تقديم مشورة للمرضى

Rational : الأساسي

- professional in Drugs
- Communicates with:
  - Patients + healthcare professional
- access drug database
- Recognize / Report side effects

4) Th Role of pharmacist in:

من باب الاختصار وحفظ المستويات:

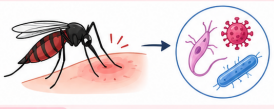
Primordial = (0)
primary = (1)
secondary = (2)
Tertiary = (3)

	For the Drug:	For the Patients:	For the Society:	For Healthy People:
Vector-born Disease	- Medicin provider for scabies / Leishmania during disasters => [2] - Provide drugs to infected areas + ensure drug distribution [2+3]	- Identify symptomatic cases + Direct to diagnostic centers and avoid delayed diagnosis [2] - Ensure Rational use of prescribed medicines [2]	- Counsel travelers + protection techniques	- Campaigns / Exhibitions [0] ex: Killing mosquito / Killing snail - Consumer health information + IEC materials [0]
Viral Infection	- Provide / distribute meals [2+3] - Rational / safe use [2] - Decrease resource waste [0+1]	- Isolation → [1] - Information → [2+3] - Diagnostic referral + symptomatic treatment [2]	- Campaigns - exhibitions [0]	- Vaccination / prevention + Counseling [0+1]
Bacterial Infection	- Provide / distribute meals [2+3] - Rational / safe use [2] - Decrease resource waste [0+1]	- Isolation → [1]+[2] - Information → [2+3] - Diagnostic referral [2]	- Antimicrobial stewardship - prevalence survey - economic issues [0+1]	- Vaccination / prevention + Counseling [0+1]
TB = Tuberculosis Infection	- Medicine distribution [2]	- Isolation → [1]+[2] - Information → [2+3] - Diagnostic referral [2] - DOTS [2]	- prevalence survey - economic issues [0+1]	- Vaccination / prevention + Counseling [0+1]
Disastar Management	- Mobile / emergency Pharmacy [2] - Drug Distribution [2] - review supply / dose [2]	- First aid [2] ex. cuts, burns	Prevalence / health survey [0+1]	—
De-addiction	- Tobacco patches / gums [2+3]	antidote dose adjusment	Health campaigns [0]	—
Maternal health	- Vaccination - Supplements - hormonal therapy 0, 1, 2, 3	- lactation / hygiene / immunization / pregnancy-medication questions [1+2+2+3] - breast-feeding [0+1]	Health campaigns [0+1]	—
Global health	up-to-date	- Drug-drug - Drug-Food - Drug-Disease interactions [1+2] - Disease education → [2+3]	- Campaigns - Cost-analysis - Pharmacovigilance [0+1]	—
Chronic Disease	- Medication compliance [2+3] - evidence-based management [1+2]	- Lifestyle modification [1+2] - routine examinations [2+3] - self checks [1+2]	- Campaigns - early dignosis / treatment / rehab - pharmacovigilance - BP / diabetes screening [0+1]	—

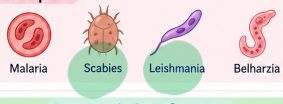
Definitions: تعريفات

Vector-Borne Diseases

Definition:
Disease caused by parasites, viruses or bacteria transmitted by mosquitoes.

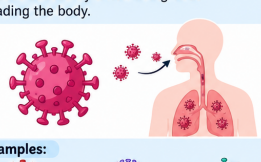


Examples:



Viral Infection

Definition:
Disease caused by a non-living virus invading the body.

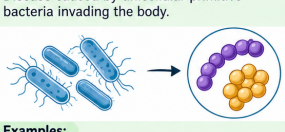


Examples:

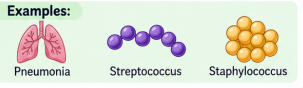


Bacterial Infection

Definition:
Disease caused by unicellular primitive bacteria invading the body.

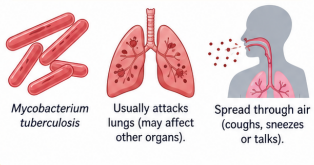


Examples:



Tuberculosis

- TB = caused by *Mycobacterium tuberculosis*.
- Usually attacks lungs, may affect other organs.
- Spread through air when a person with TB coughs, sneezes or talks.



اختصرها بـ:

Keyword	Level
Risk factors / vector control	P0
Campaigns / prevalence / cost / pharmacovigilance	P0 + P1
Vaccination / prevention	P0 + P1
Traveler protection	P1
Diagnosis / symptomatic treatment / first aid / DOTS	P2
Rational & safe use	P2
Drug distribution / patient information	P2 + P3
Lifestyle / self-checks / evidence-based management	P1 + P2
Long-term / compliance	P2 + P3

The role of pharmacist:

Levels:↓
Primordial = (0)
primary = (1)
secondary = (2)
Tertiary = (3)

- vector born : - For drug: Medicin provider → -oral : scabies _ Leishmania Level : 2
  - For society : Counsel travelers
  - For healthy people: holding campaigns and Exhibitions Level : 0

- viral Infection : - For society : holding campaigns and Exhibitions Level : 0
  - For healthy people: vaccination /prevention

- TB = Tuberculosis Infection : - For patients: DOTS (Directly Observed Treatments) Level : 2

- Disaster Management: - For drug: Mobile pharmacy (e prescription and apps) Level : 2
  - For patients: proride first aid treatment → cuts , burns → Level : 2

- De-addiction : - For patients: anti-dots adjustment
  - For Society: Health campaigns → Level : 0

- Maternal health : - For drug : vaccination
  - For patients: Answering all questions (immunization, medications during pregnancy and lactation)

- Global health : - For drug : up-to-date information
  - For society : cost-analysis

- General notice in role of pharmacist in public Health

* Pharmacist public health role:

Epidemiology → Awareness → Patient counseling

(studying)

Epidemiology: study of distribution and prevention of health-related events in a specific population.

Patient counseling: important role that a pharmacist provides

→ Measurements → Publications

Measurements:
- Blood pressure
- Blood sugar
- weight
- Height

Publications:
- Books
- Pamphlets
- Patient Information Leaflets

Pharmacy Job Opportunities

1. COMMUNITY PHARMACY

Major Role: Dispense prescriptions + OTC for minor ailments

- Counsel patients
- Promote public health: smoking, weight control, lifestyle, BP
- Provide cosmetics + household items
- Most community pharmacists own their own business
- Business management education is essential: Law, Marketing, Accounting, Pharmacy Administration
- Drawback: Long working hours + few supporting staff



Dispense → Counsel → Public Health → Business

2. TELEPHARMACY

Telepharmacy = pharmacist follows up the dispensing process at multiple locations.

Technician	receives order	enters into computer	prepares product (product selection + labeling + billing)	Pharmacist
				FINAL CHECK

Video + Image + Audio links

Video | Image | Audio

Patient counseling through video

Technician prepares → Pharmacist checks → Patient video counseling

3. HOSPITAL PHARMACY



- Calculates IV dosage
- Dispenses for in-patients
- Manages drug distribution

IV – Inpatient – Distribution

Specializations:

- Nuclear pharmacy
- Parenteral medication
- Poisoning center
- Clinical trials



4. CLINICAL PHARMACY

Clinical Pharmacist → TDM + DIC + Patient Counseling

- TDM = Therapeutic Drug Monitoring
- DIC = Drug Information Center
- Should hold PharmD
- Develop pharmaceutical care plan for each patient
- Works in hospital with patient + healthcare professions

Monitor (TDM) | Information (DIC) | Patient Counseling | Care Plan

Clinical = Monitor + Information + Patient Counseling

TDM → DIC → Counseling → Care Plan

5. HOSPITAL vs CLINICAL PHARMACY

Hospital Pharmacy	Clinical Pharmacy
IV dosage	TDM
In-patient dispensing	DIC
Drug distribution	Patient counseling
Hospital drug operations	Individual pharmaceutical care plan

Hospital = Drug distribution/system

Clinical = Patient therapy/care

6. INDUSTRIAL PHARMACY

A) R&D

- Develop new drugs
- New dosage forms
- Solve manufacturing problems
- Requires strong Pharmaceutics knowledge

B) Manufacturing

- Cost-saving
- High-efficient techniques
- Safety and health of workers

C) QC → quality control

- Analysis of drug in dosage form
- Raw materials
- Degradation products

Develop → Produce → Analyze

Thanx to myself and Dr guiding and ChatGPT helping